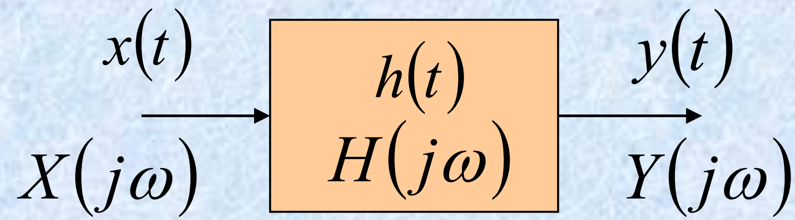
Chapter 6

Time and Frequency Characterization of Signals and Systems

Time-Domain:

$$y(t) = x(t) * h(t)$$



Frequency-Domain:

$$Y(j\omega) = X(j\omega)H(j\omega)$$

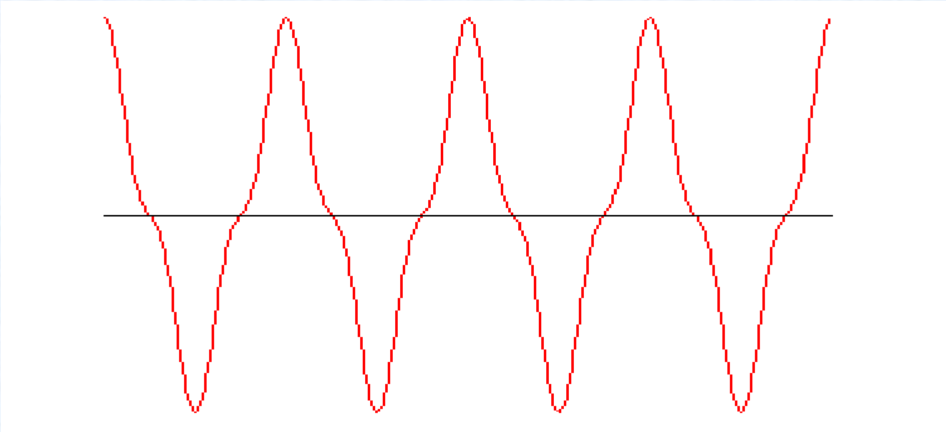
§6.1 The Magnitude-Phase Representation （幅度-相位） of the Fourier Transform

$$x(t) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\omega)e^{j\omega t}d\omega$$

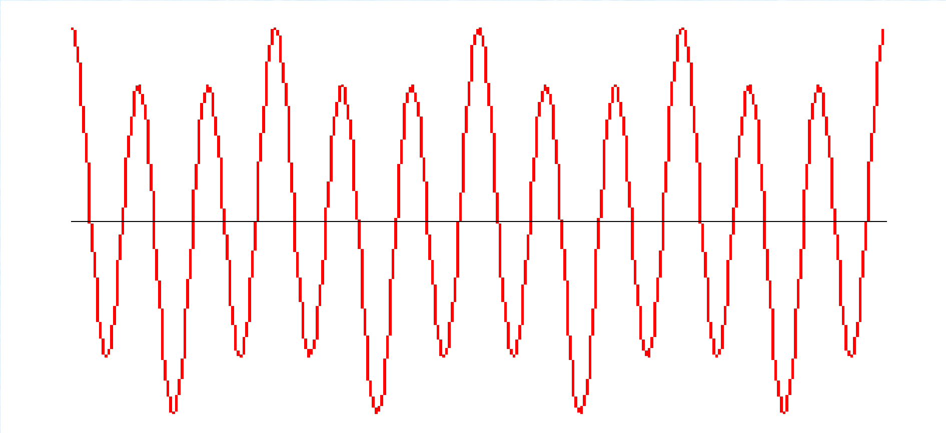
$$X(j\omega) = |X(j\omega)|e^{j\angle X(j\omega)}$$

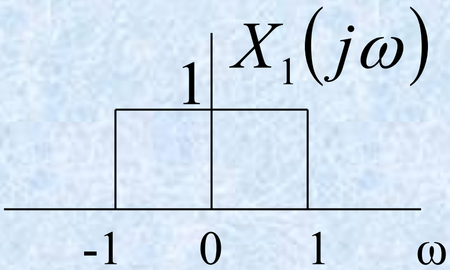
$|X(j\omega)|$ ——**Magnitude** $|X(j\omega)|^2$ ——**Energy-Density**

$$x_1(t) = 4\cos 2\pi t + \cos 6\pi t$$

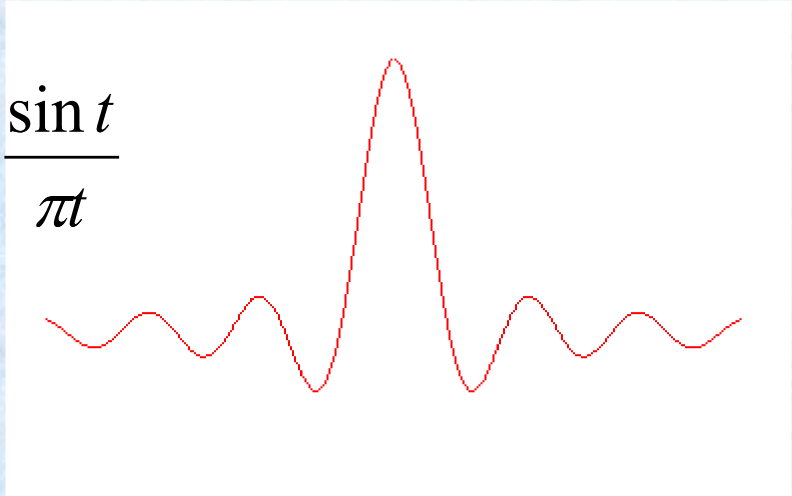


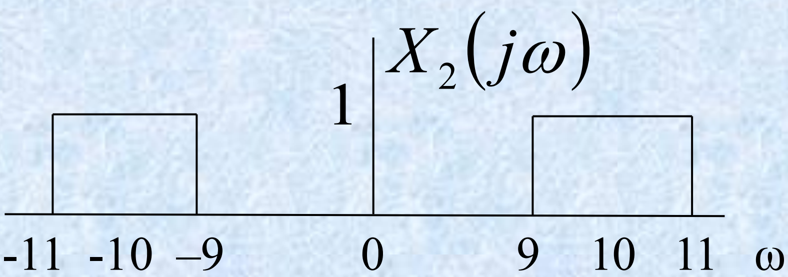
$$x_2(t) = \cos 2\pi t + 4\cos 6\pi t$$

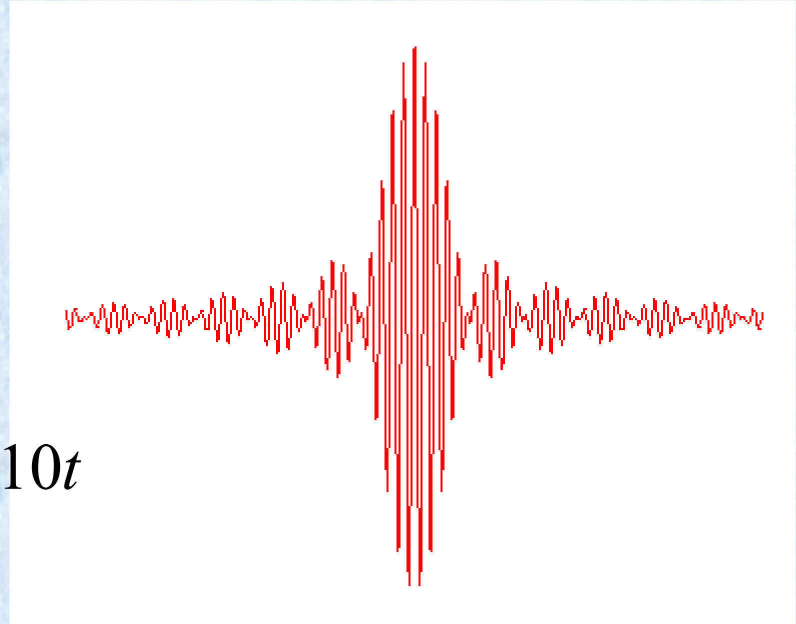




$$x_1(t)=\frac{\sin t}{\pi t}$$

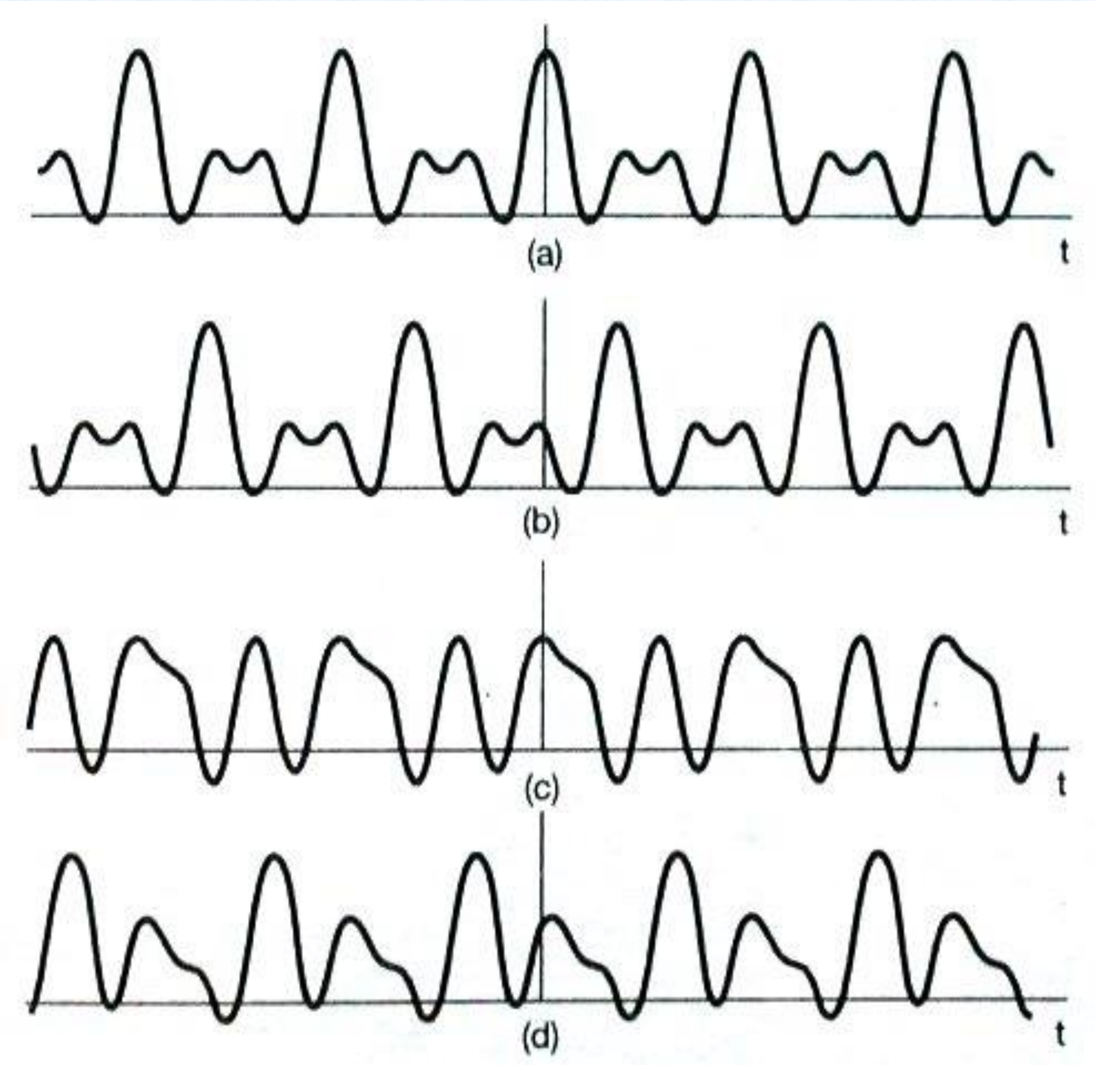






$$x_2(t)=2\frac{\sin t}{\pi t}\cos 10t$$

$$x(t) = 1 + \frac{1}{2}\cos(2\pi t + \Phi_1) + \cos(4\pi t + \Phi_2) + \frac{2}{3}\cos(6\pi t + \Phi_3)$$



$$\Phi_1 = \Phi_2 = \Phi_3 = 0$$

$$\Phi_1 = 4, \Phi_2 = 8, \Phi_3 = 12$$

$$\Phi_1 = 6, \Phi_2 = -2.7, \Phi_3 = 0.93$$

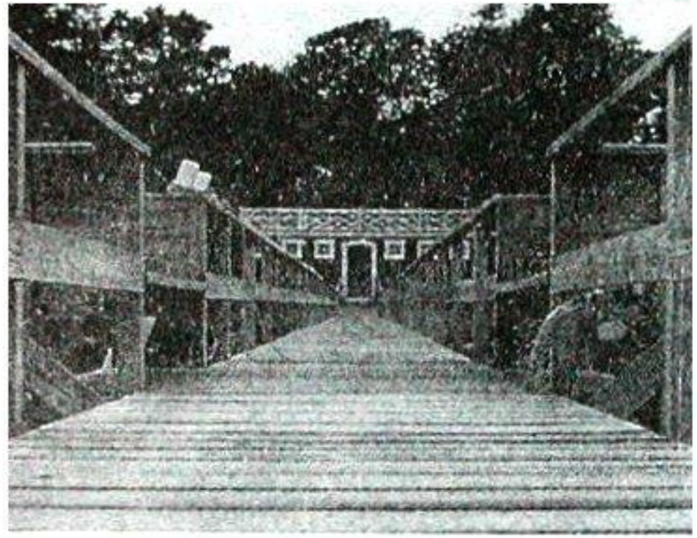
$$\Phi_1 = 1.2, \Phi_2 = 4.1, \Phi_3 = -7.2$$

2. A black-and-white picture

$$x(t_1,t_2)\overset{F}{\longleftrightarrow}X(j\omega_1,j\omega_2)$$

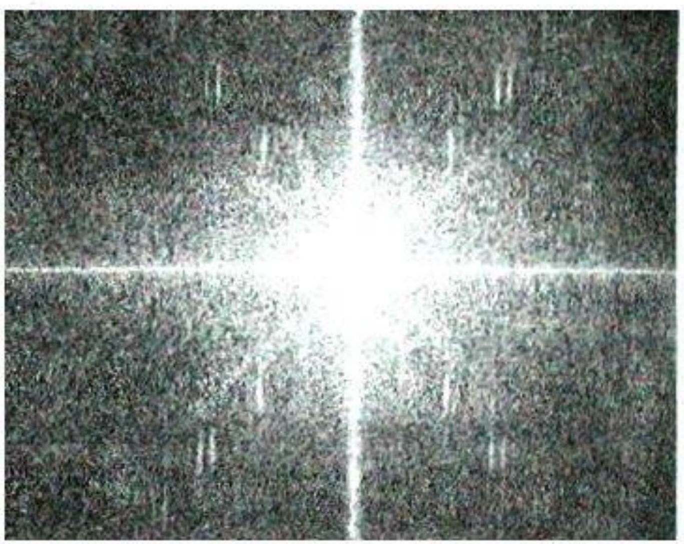
$$X(j\omega_1,j\omega_2)=\int_{-\infty}^{+\infty}x(t_1,t_2)e^{-j\omega_1 t_1}e^{-j\omega_2 t_2}dt_1 dt_2$$

Figure 6.2



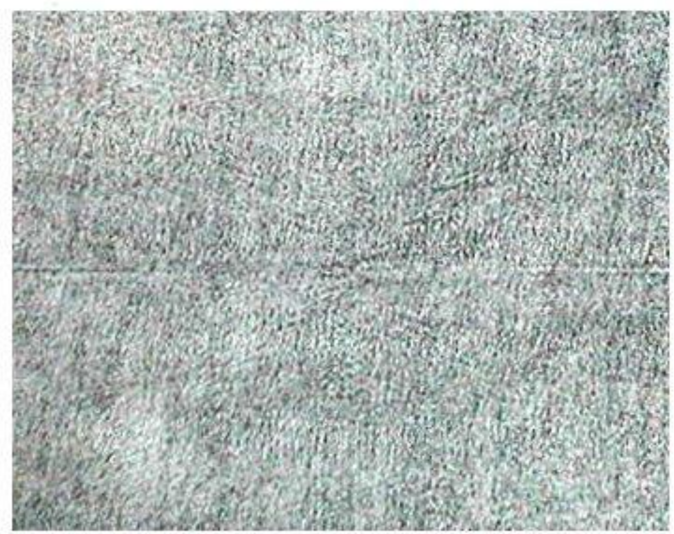
(a)

$$|X(j\omega_1, j\omega_2)|$$

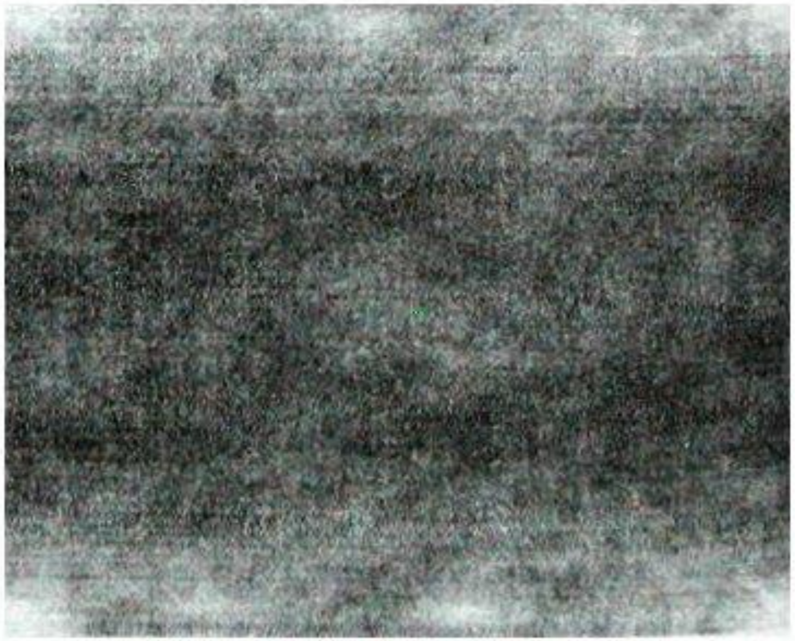


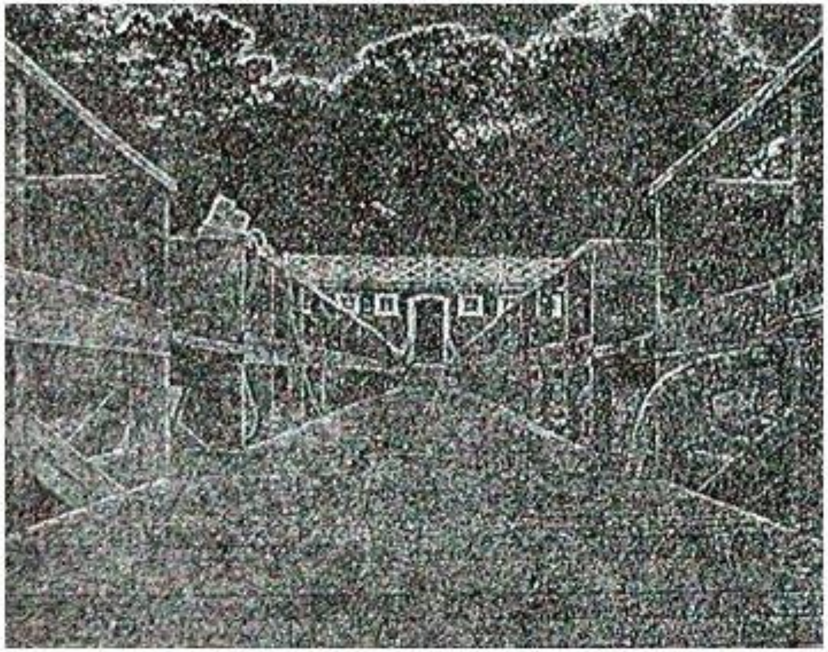
(b)

$$\angle X(j\omega_1, j\omega_2)$$



(c)


(d)


(e)

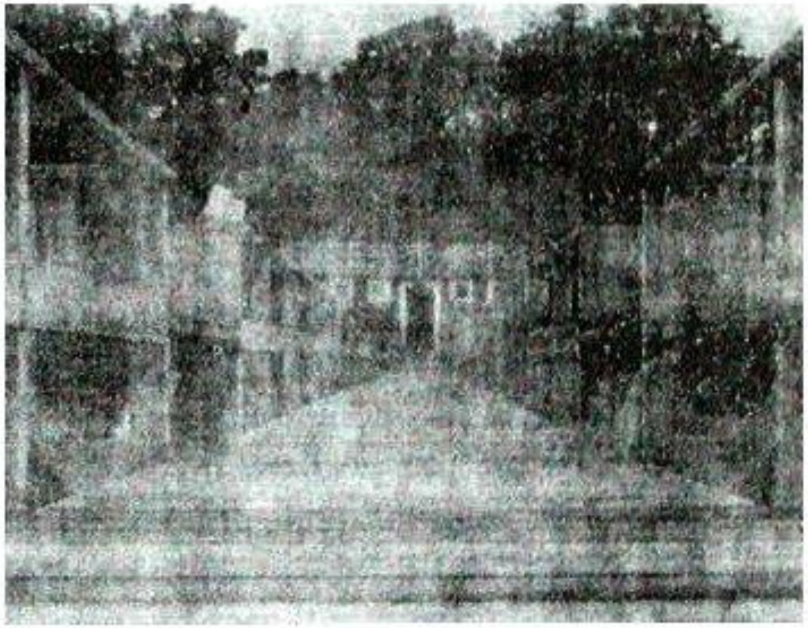
$$Magnitude : | X(j\omega_1, j\omega_2) |$$

$$Phase : \angle X(j\omega_1, j\omega_2) = 0$$

$$Magnitude : | X(j\omega_1, j\omega_2) | = 1$$

$$Phase : \angle X(j\omega_1, j\omega_2)$$

(g)



(f)

$$\textbf{\textit{Magnitude}} \;+\; \textbf{\textit{Phase}}: \angle X(j\omega_1, j\omega_2)$$

§6.2 The Magnitude-Phase Representation of the Frequency Response of LTI Systems

$$Y(j\omega) = X(j\omega)H(j\omega)$$

$$|Y(j\omega)| = |X(j\omega)||H(j\omega)|$$

$$\angle Y(j\omega) = \angle X(j\omega) + \angle H(j\omega)$$

$|H(j\omega)|$ ——Gain

$\angle H(j\omega)$ ——Phase Shift

§6.2.1 Linear and Nonlinear Phase

1. Linear Phase

$$H(j\omega) = e^{-j\omega t_0} \qquad |H(j\omega)| = 1 \qquad \angle H(j\omega) = -\omega t_0$$

$$\angle Y(j\omega) = \angle X(j\omega) - \omega t_0$$

$$x(t)=\cos\left(t/\sqrt{3}\right)+\cos t+\cos\sqrt{3}t$$

$$H(j\omega)=e^{-j\omega} \qquad \left|H(j\omega)\right|=1 \qquad \angle H(j\omega)=-\omega$$

$$\cos\left(t/\sqrt{3}\right)\rightarrow\frac{1}{2}e^{jt/\sqrt{3}}e^{-j/\sqrt{3}}+\frac{1}{2}e^{-jt/\sqrt{3}}e^{j/\sqrt{3}}=\cos\left((t-1)/\sqrt{3}\right)$$

$$\cos(t)\rightarrow\frac{1}{2}e^{jt}e^{-j}+\frac{1}{2}e^{-jt}e^{j}=\cos(t-1)$$

$$\cos\left(\sqrt{3}t\right)\rightarrow\frac{1}{2}e^{j\sqrt{3}t}e^{-j\sqrt{3}}+\frac{1}{2}e^{-j\sqrt{3}t}e^{j\sqrt{3}}=\cos\left(\sqrt{3}(t-1)\right)$$

$$y(t)=\cos\left((t-1)/\sqrt{3}\right)+\cos(t-1)+\cos\sqrt{3}(t-1)$$

2. Nonlinear Phase

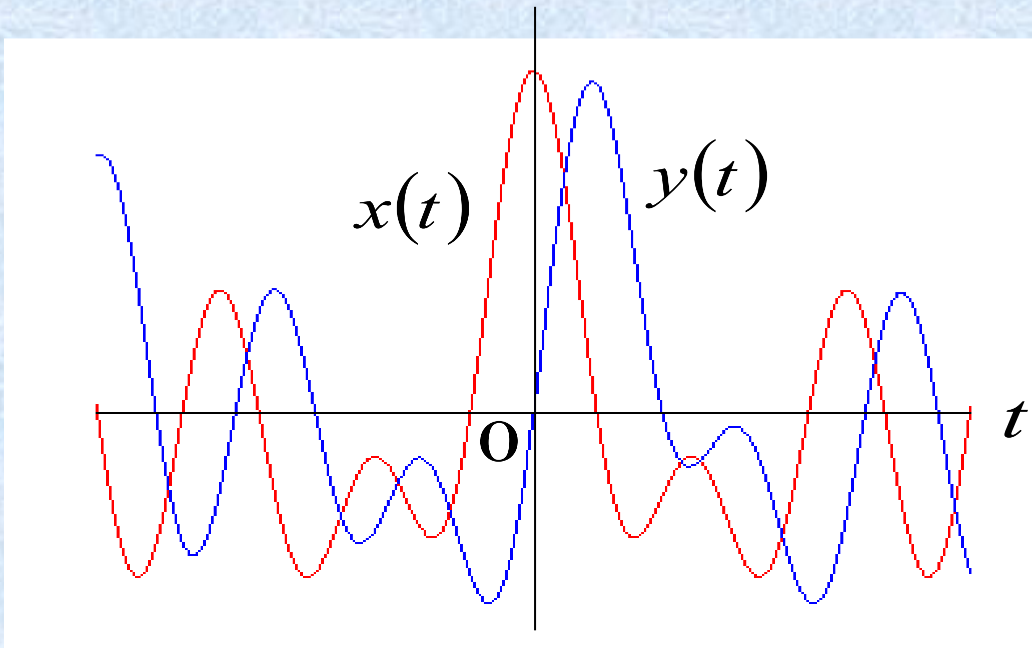
$$H(j\omega)=\frac{1-j\omega}{1+j\omega} \qquad \left|H(j\omega)\right|=1 \qquad \angle H(j\omega)=-2tg^{-1}\omega$$

$$\cos\left(t/\sqrt{3}\right) \to \frac{1}{2}e^{jt/\sqrt{3}}e^{\underbrace{-j2arctg\left(\sqrt{3}/3\right)}_{-\pi/3}} + \frac{1}{2}e^{-jt/\sqrt{3}}e^{\underbrace{j2arctg\left(\sqrt{3}/3\right)}_{\pi/3}}$$

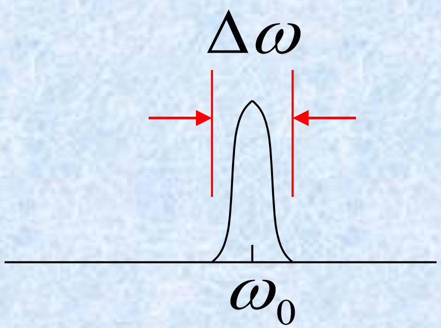
$$= \cos\left(t/\sqrt{3} - \pi/3\right)$$

$$y(t) = \cos\left(t/\sqrt{3} - \pi/3\right) + \cos\left(t - \pi/2\right) + \cos\left(\sqrt{3}t - 2\pi/3\right)$$

$$y(t) = \cos\left(\left(t - \underline{\pi/\sqrt{3}}\right)/\sqrt{3}\right) + \cos\left(t - \underline{\pi/2}\right) + \cos\left(\sqrt{3}\left(t - \underline{2\pi/\left(3\sqrt{3}\right)}\right)\right)$$



§6.2.2 Group delay （群延时）



Consider a <u>narrowband signal</u> 窄带信号 $\Delta\omega << \omega_0$

$$X(j\omega)=\begin{cases}\neq 0 & |\omega-\omega_0|<\dfrac{\Delta\omega}{2}\\ 0 & |\omega-\omega_0|>\dfrac{\Delta\omega}{2}\end{cases}$$

$$\angle H(j\omega)=-\phi-\omega\alpha$$

$$\boxed{Y(j\omega)=X(j\omega)|H(j\omega)|e^{-j\phi}e^{-j\omega\alpha}}$$

α——Group delay at ω=ω_0

The Group Delay at each frequency

$$\boxed{\tau(\omega)=-\frac{d}{d\omega}\{\angle H(j\omega)\}}$$

Example: $$y(t) = x(t - t_0)$$

$$H(j\omega) = e^{-j\omega t_0}$$

$$\angle H(j\omega) = -\omega t_0$$

$$\tau(\omega) = t_0 \quad (signal\ delay)$$

Example 6.1 Consider an all-pass system.

$$H(j\omega)=\prod_{\mathrm{i}=1}^{3} H_{\mathrm{i}}(j\omega)$$

$$H_{\mathrm{i}}(j\omega)=\frac{1+(j\omega/\omega_i)^2-2j\zeta_i(\omega/\omega_i)}{1+(j\omega/\omega_i)^2+2j\zeta_i(\omega/\omega_i)}$$

$$\left|H_{\mathrm{i}}(j\omega)\right|=1\Rightarrow\left|H(j\omega)\right|=1$$

$$\angle H_{\mathrm{i}}(j\omega)=-2tg^{-1}\left\{\frac{2\zeta_i(\omega/\omega_i)}{1-(\omega/\omega_i)^2}\right\}$$

$$\angle H(j\omega)=\sum_{\mathrm{i}=1}^{3}\angle H_{\mathrm{i}}(j\omega)$$

$$\omega/\omega_i=f/f_i$$

Let $\begin{cases} f_1=50Hz \\ f_2=150Hz \\ f_3=300Hz \end{cases}$ $\begin{cases} \zeta_1=0.066 \\ \zeta_2=0.033 \\ \zeta_3=0.058 \end{cases}$

Figure 6.5

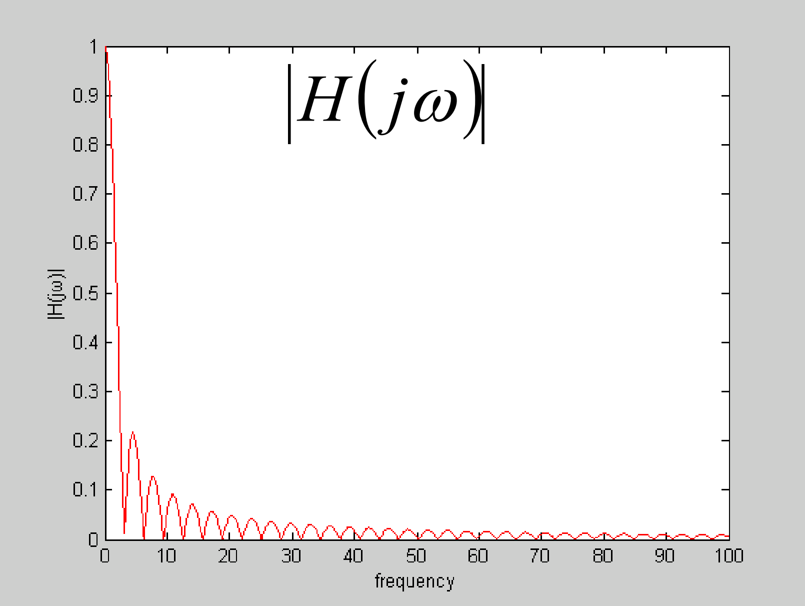
§6.2.3 Log-Magnitude and Bode Plots

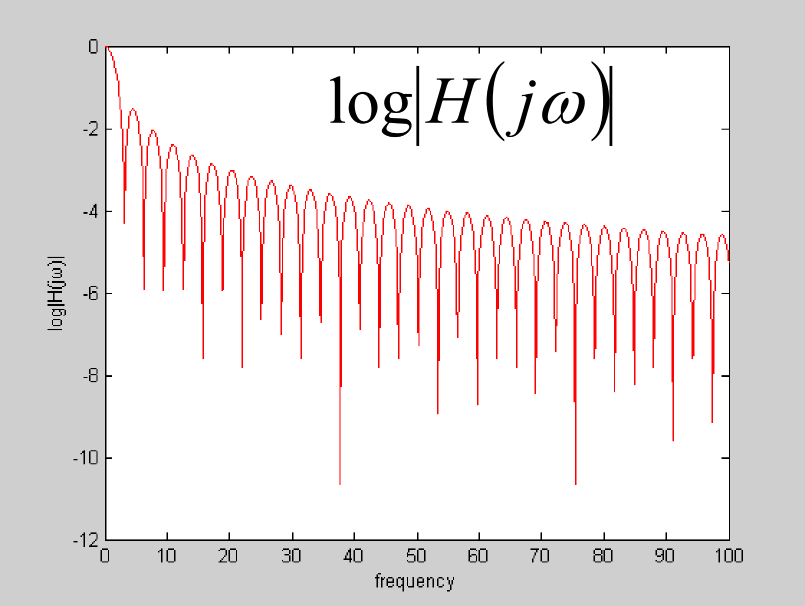
对数模和波特图

$$\log|Y(j\omega)| = \log|X(j\omega)| + \log|H(j\omega)|$$

Bode Plots：Log-Frequency——Log-Magnitude

$$H(j\omega) = \frac{\sin\omega}{\omega}$$





Example

$$\tau \frac{dy(t)}{dt} + y(t) = x(t)$$

$$H(j\omega) = \frac{1/\tau}{j\omega + 1/\tau}$$

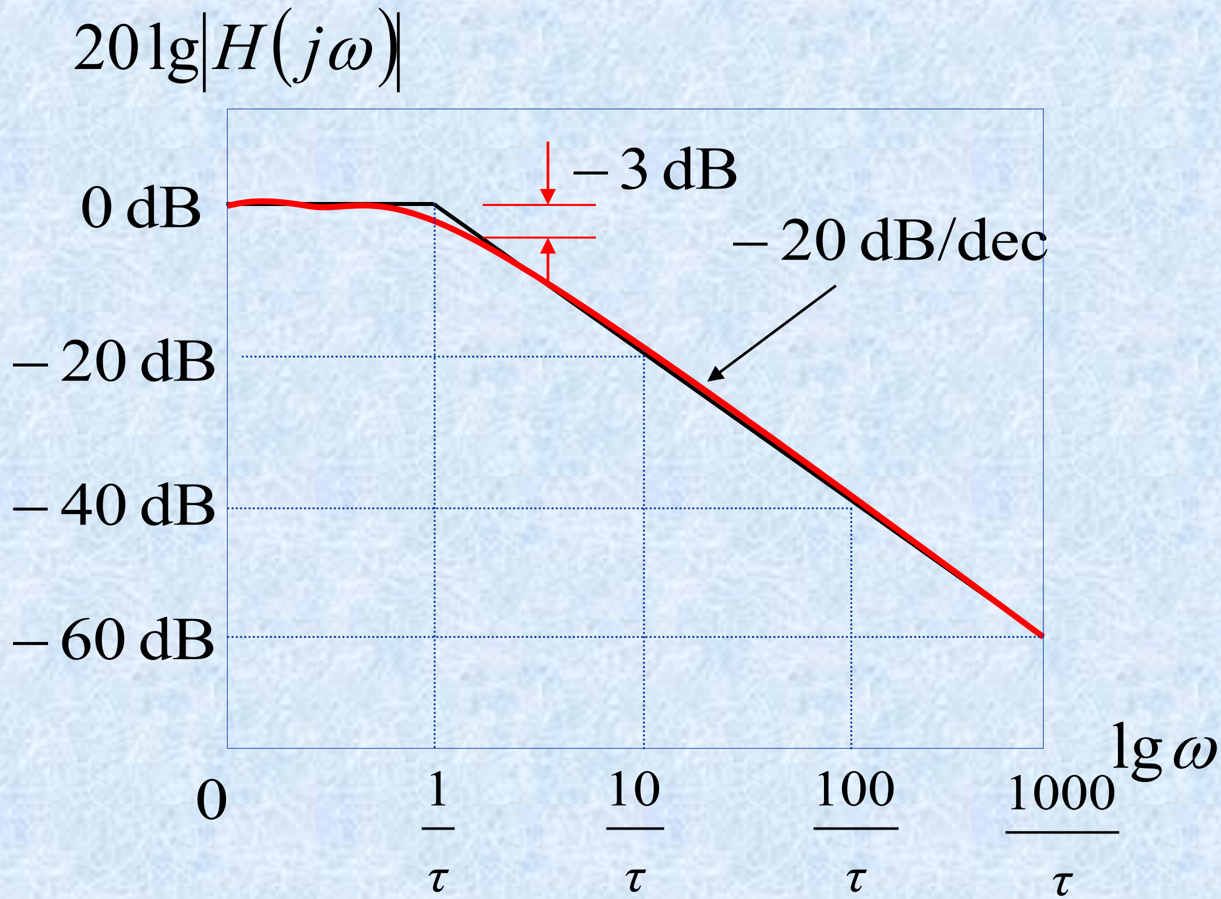
$$|H(j\omega)| = \frac{1/\tau}{\sqrt{\omega^2 + (1/\tau)^2}}$$

$$20\lg|H(j\omega)| = 20\lg\left[1/\left(\tau\sqrt{\omega^2 + (1/\tau)^2}\right)\right]$$

$$(\mathrm{a})\ \omega << 1/\tau \quad 20\lg|H(j\omega)| \approx 0$$

$$(\mathrm{b})\ \omega >> 1/\tau \quad 20\lg|H(j\omega)| \approx -20\lg\omega - 20\lg\tau$$

It is approximately a linear function of lg(ω)

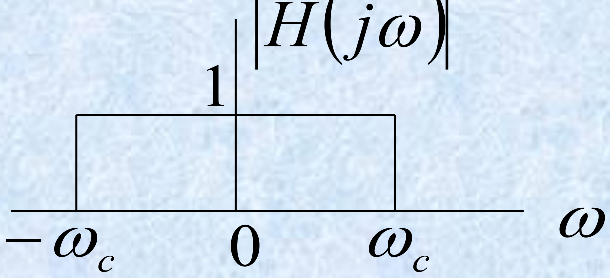


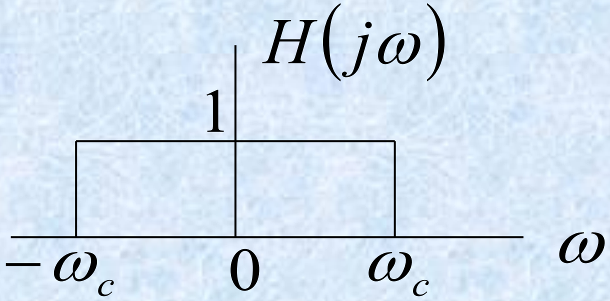
$$-20\lg\omega-20\lg\tau=0 \qquad \Rightarrow \omega=\frac{1}{\tau}$$

$$20\lg|H(j\omega)|\big|_{\omega=1/\tau}=-3\,(\text{dB})$$

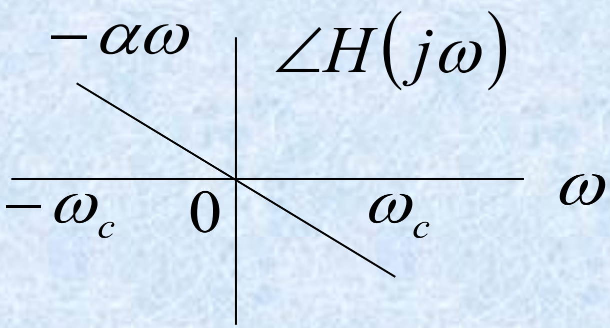
§6.3 Time-Domain Properties of ideal Frequency-Selective Filters

All-pass system with linear phase

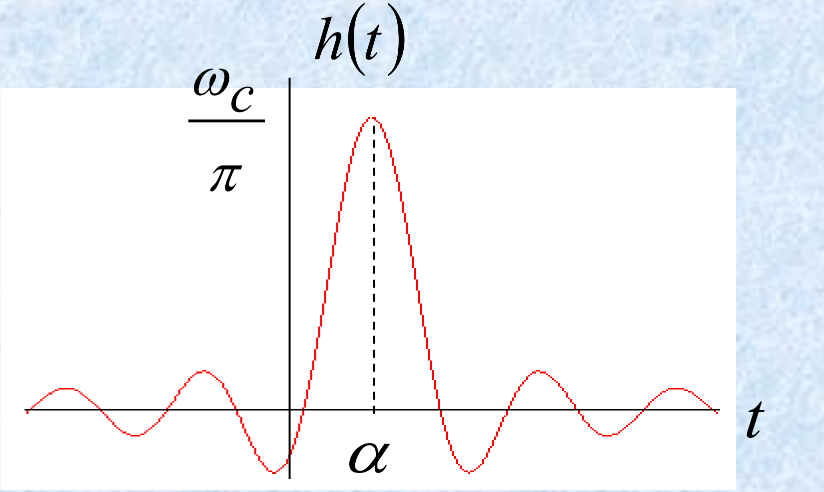




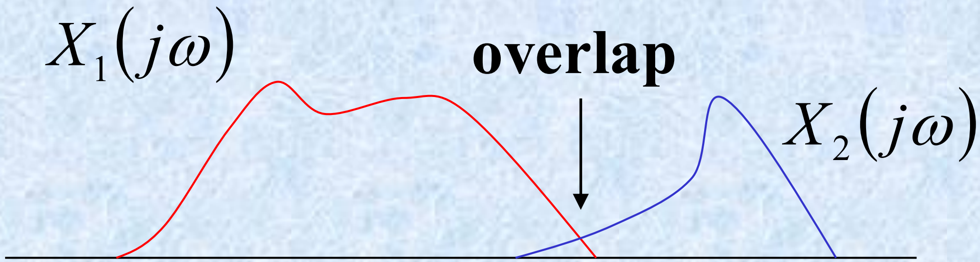
$$\angle H(j\omega) = 0$$



$$h(t) = \frac{\sin \omega_c (t-\alpha)}{\pi (t-\alpha)}$$



§6.4 Time-Domain and Frequency-Domain Aspects of Nonideal filters

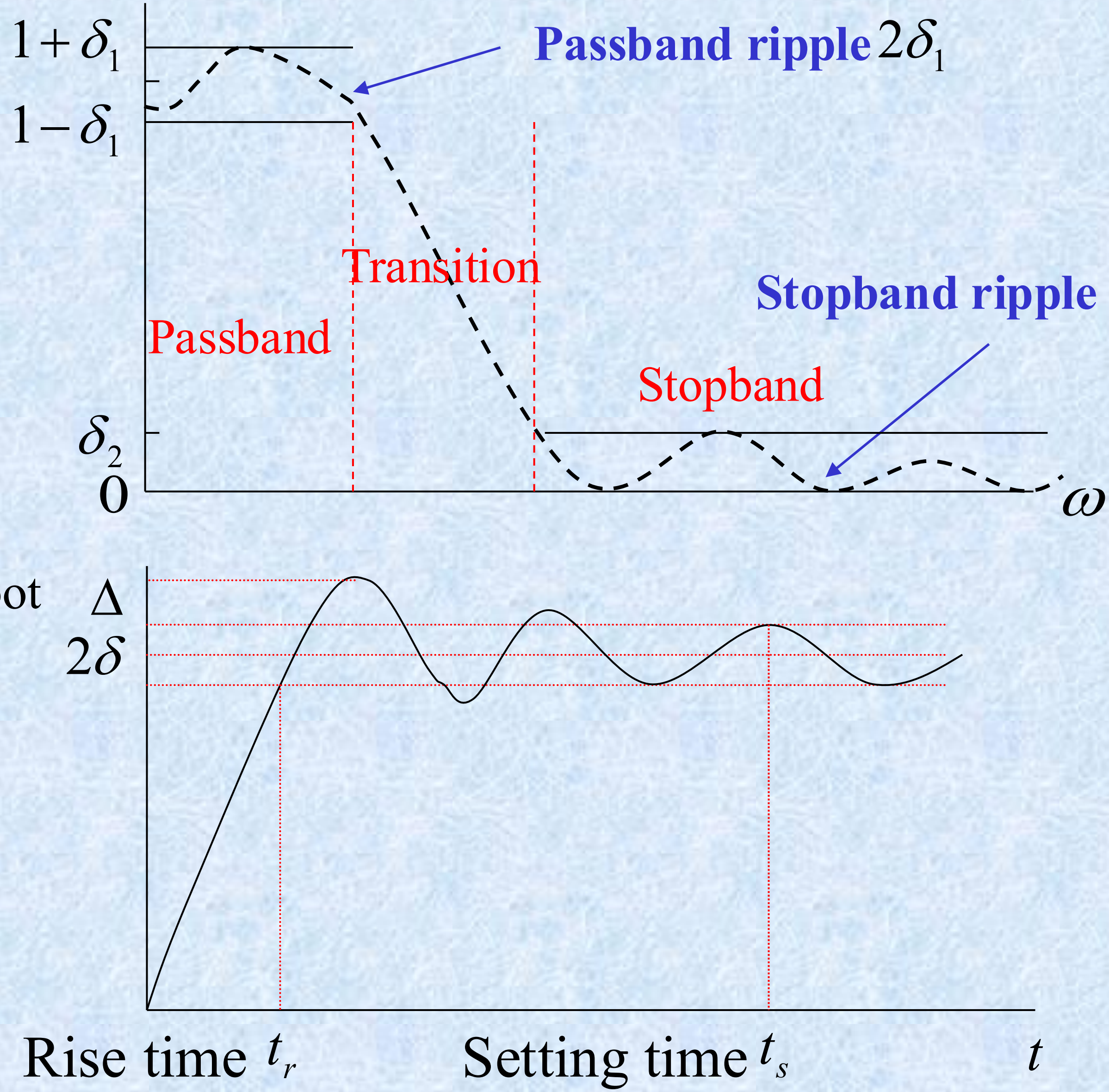


1. Gradual transition from passband to stopband.
2. The step response approaches a constant value without oscillations.
3. When filtering is to be carried out in real time, causality is a necessary constraint.
4. In many contexts, we just need a simple filter.

Nonideal Filter

Magnitude $\left|H\left(j\omega\right)\right|$

The unit response $s(t)$



Homework:

6.5 6.23 6.27